

Simple Machines & Structures

A Guide for Young Engineers
(Grade 6)



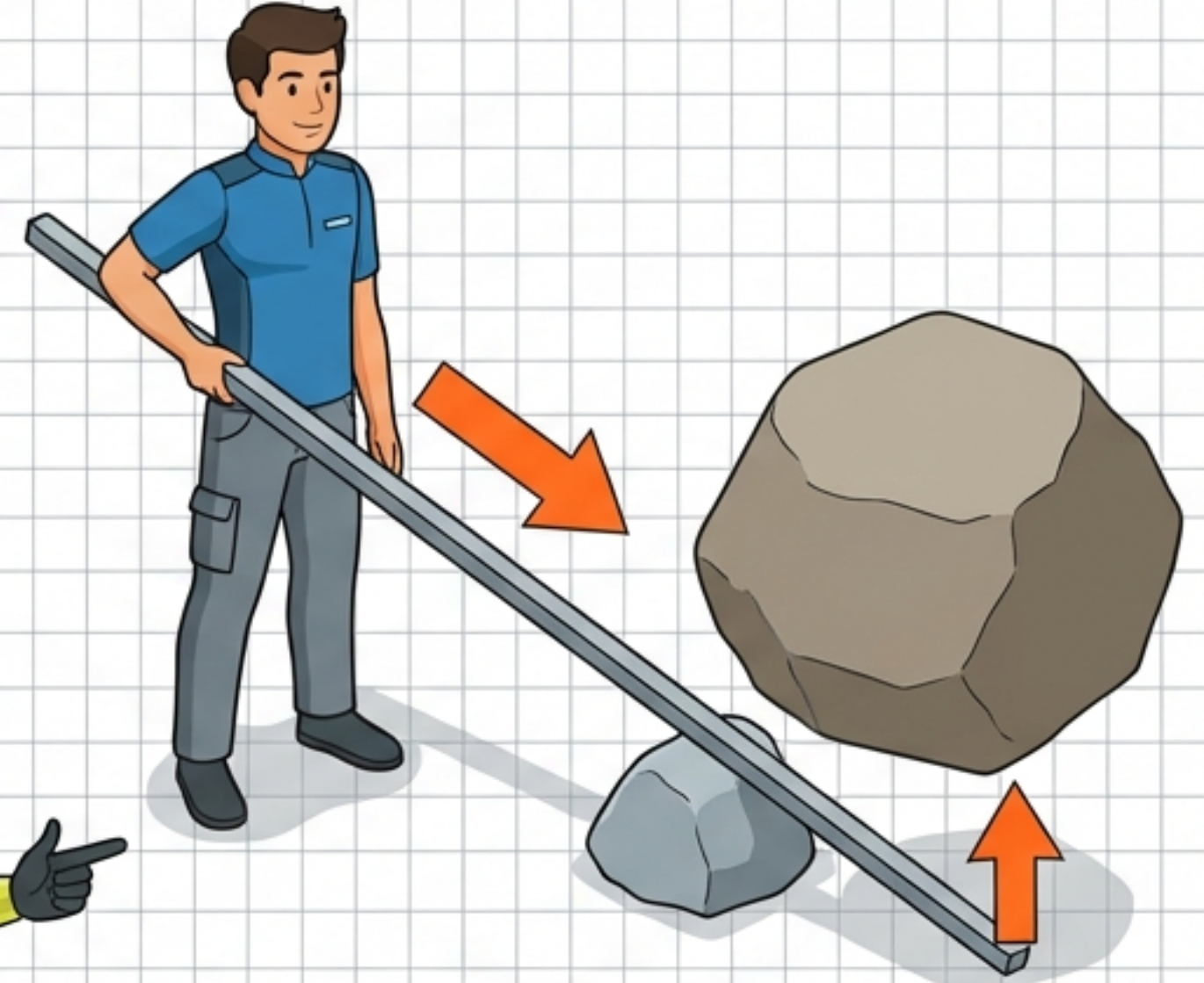
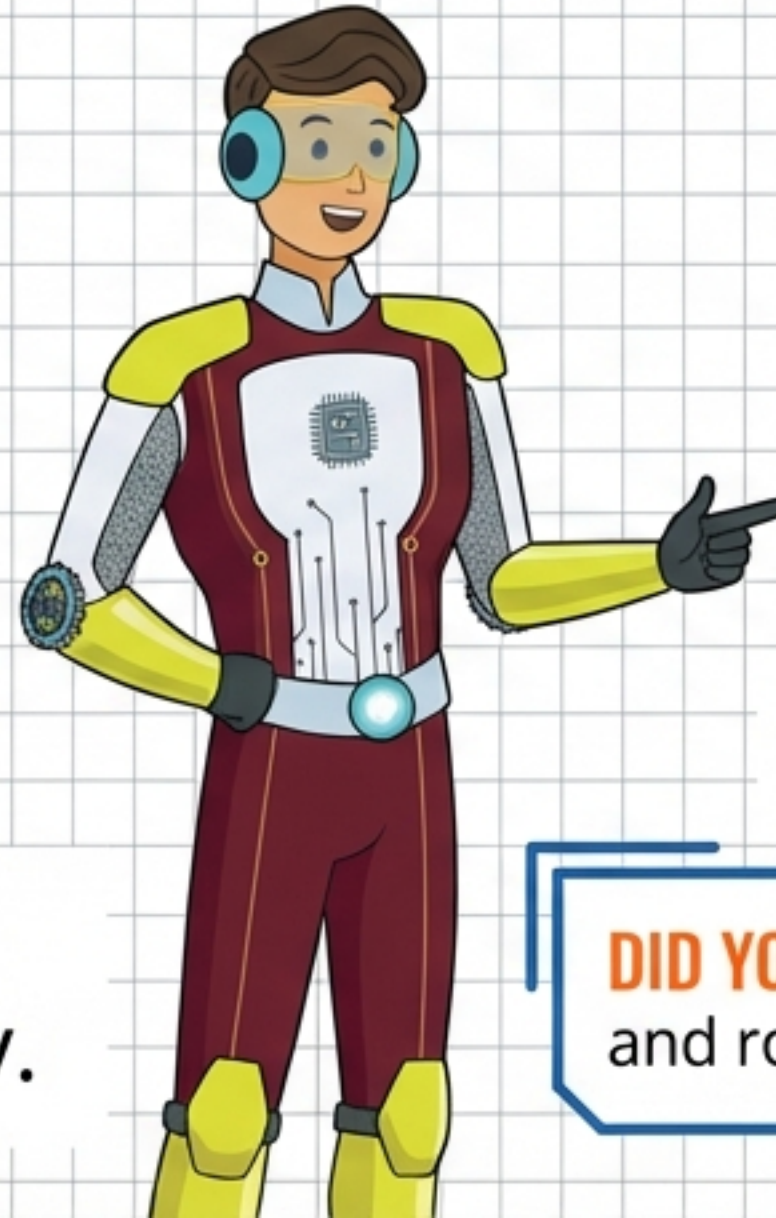
Why Do We Build Machines?



STRUGGLING WITH BARE HANDS.

Machines help us clean, move objects, create, repair, and play.

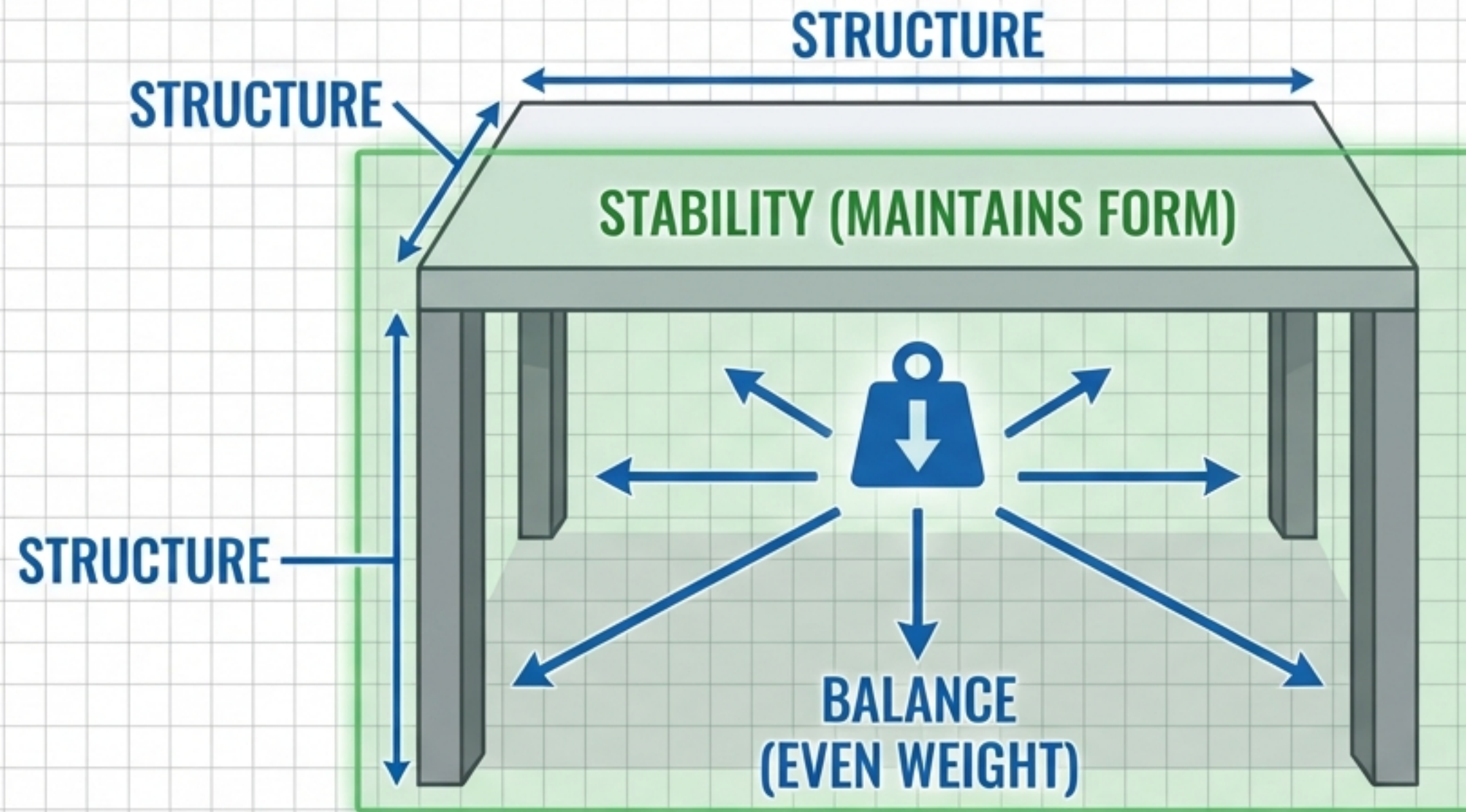
MACHINES MULTIPLY OUR FORCE & MAKE WORK EASIER!



USING A LEVER (SIMPLE MACHINE).

DID YOU KNOW? All complex machines—like cars and robots—are built from simple machines.

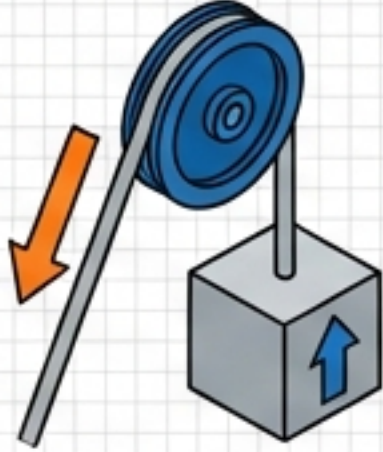
The Foundation: Structure, Stability, & Balance



Structure: Organization of parts.
Stability: Ability to not fall or break.
Balance: Even distribution of weight.

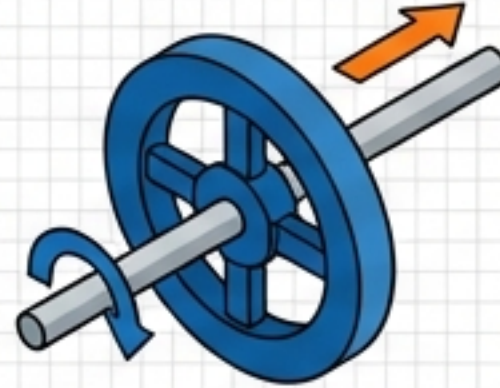


The 6 Simple Machines



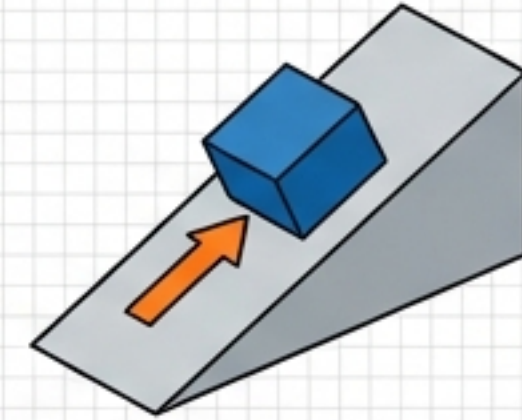
Pulley

Uses wheels and rope to lift things.



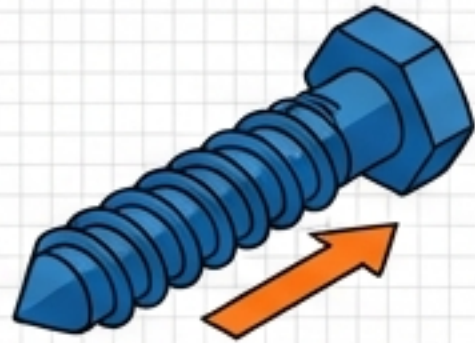
Wheel & Axle

A rod attached to a wheel to move loads.



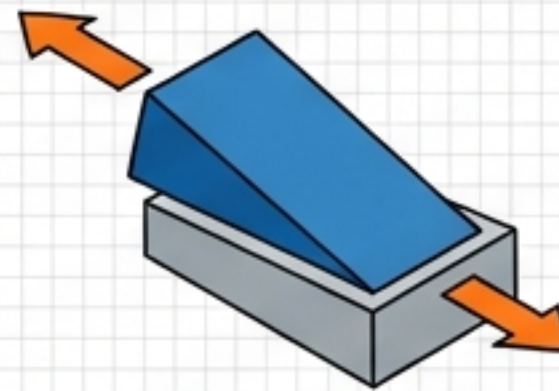
Inclined Plane

A slanted surface to move objects up or down.



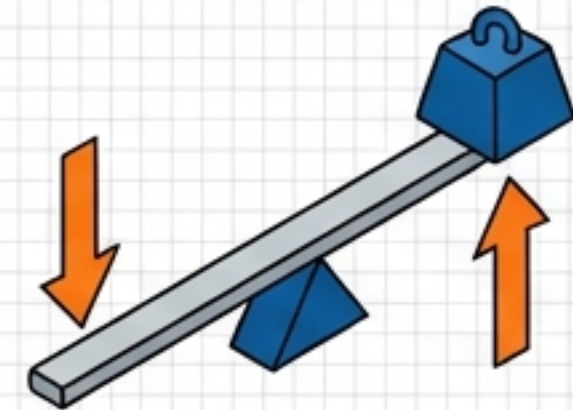
Screw

An inclined plane wrapped around a pole.



Wedge

Two inclined planes used to split or cut.

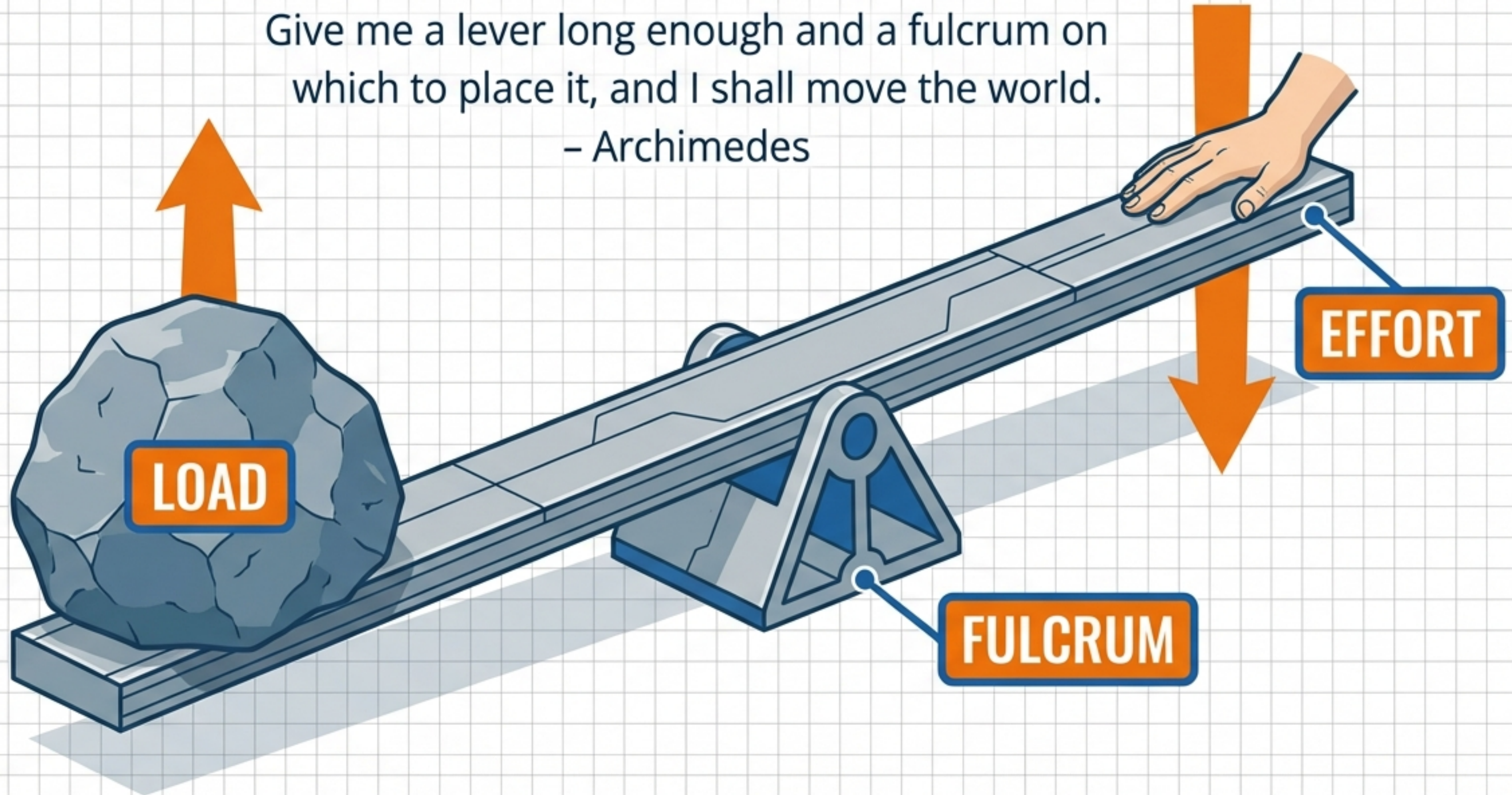


Lever

A bar resting on a fulcrum to lift weights.

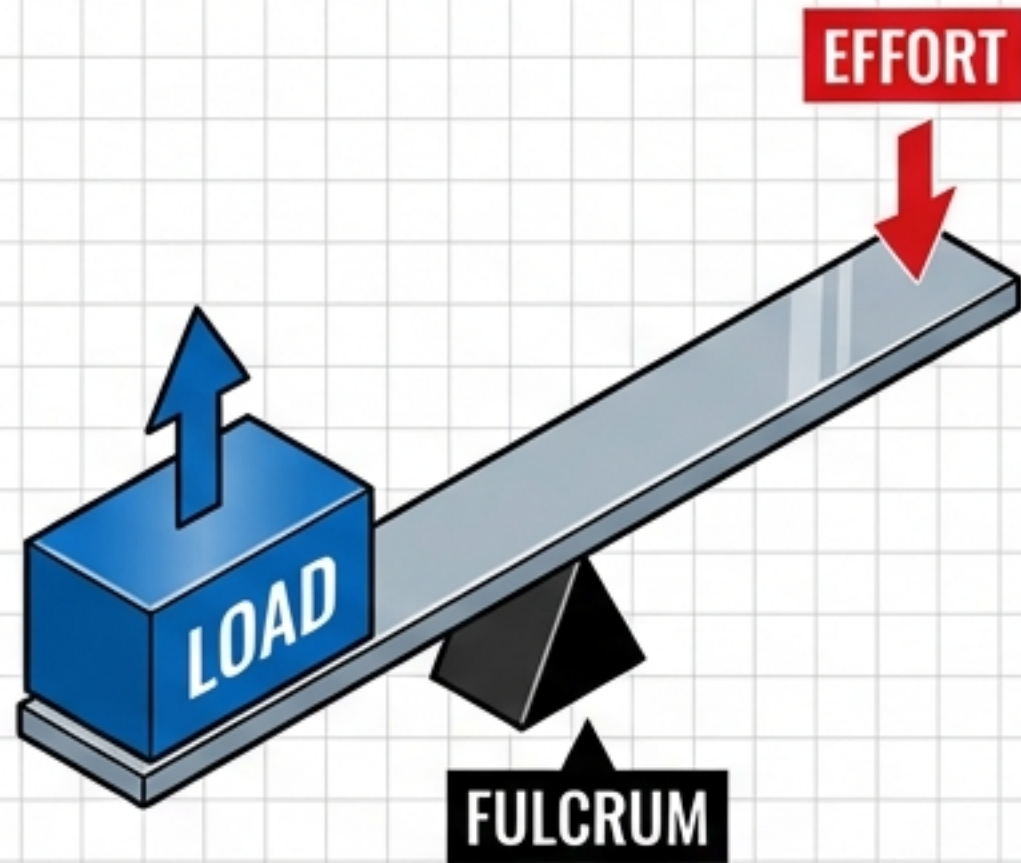
How a Lever Works

Give me a lever long enough and a fulcrum on which to place it, and I shall move the world.
– Archimedes



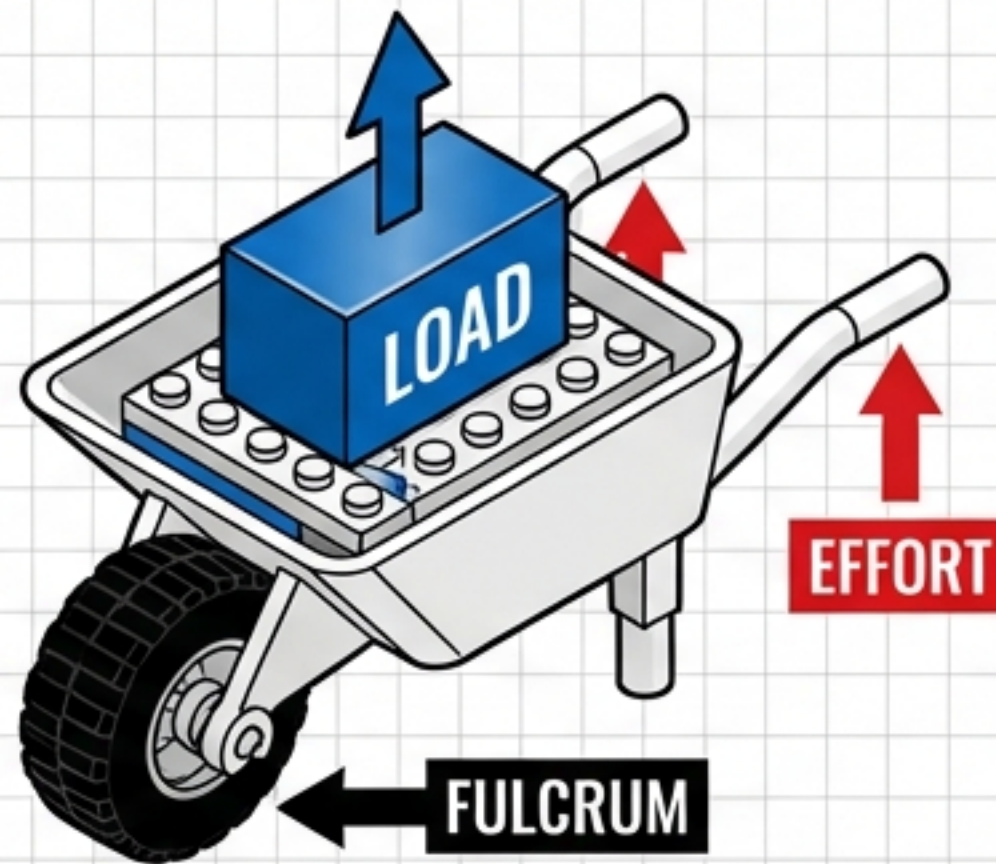
The Three Classes of Levers

Class 1



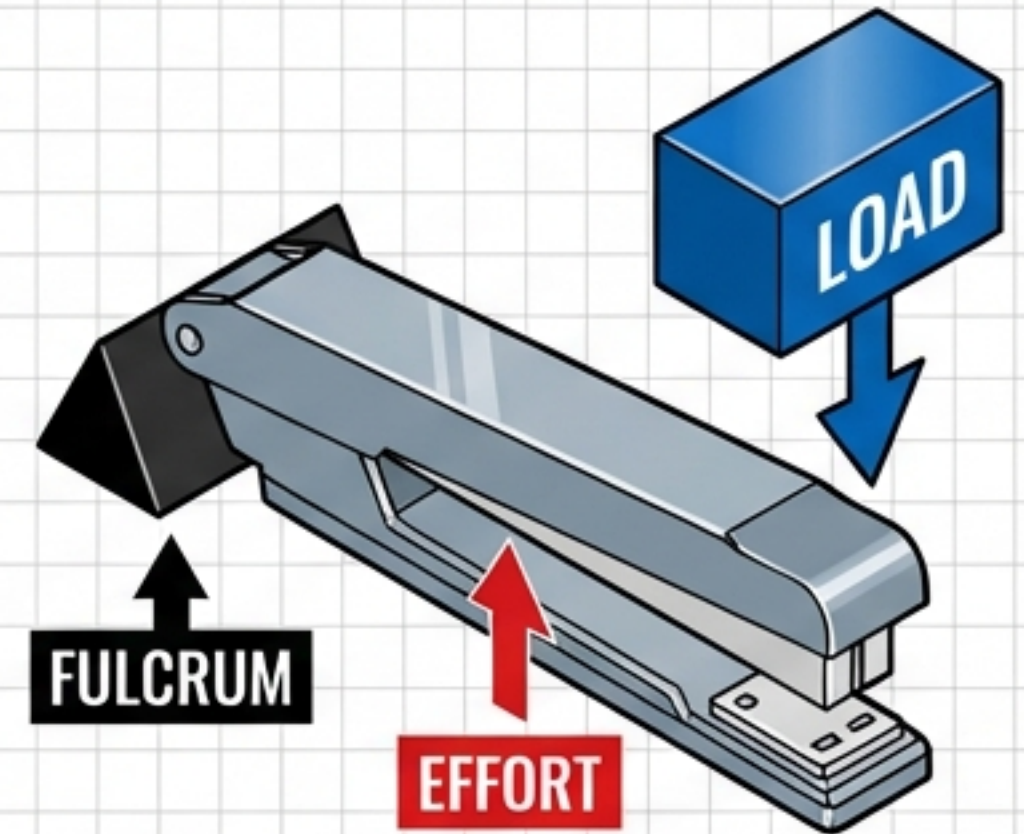
Fulcrum is between
Effort and Load.

Class 2



Load is between
Fulcrum and Effort.

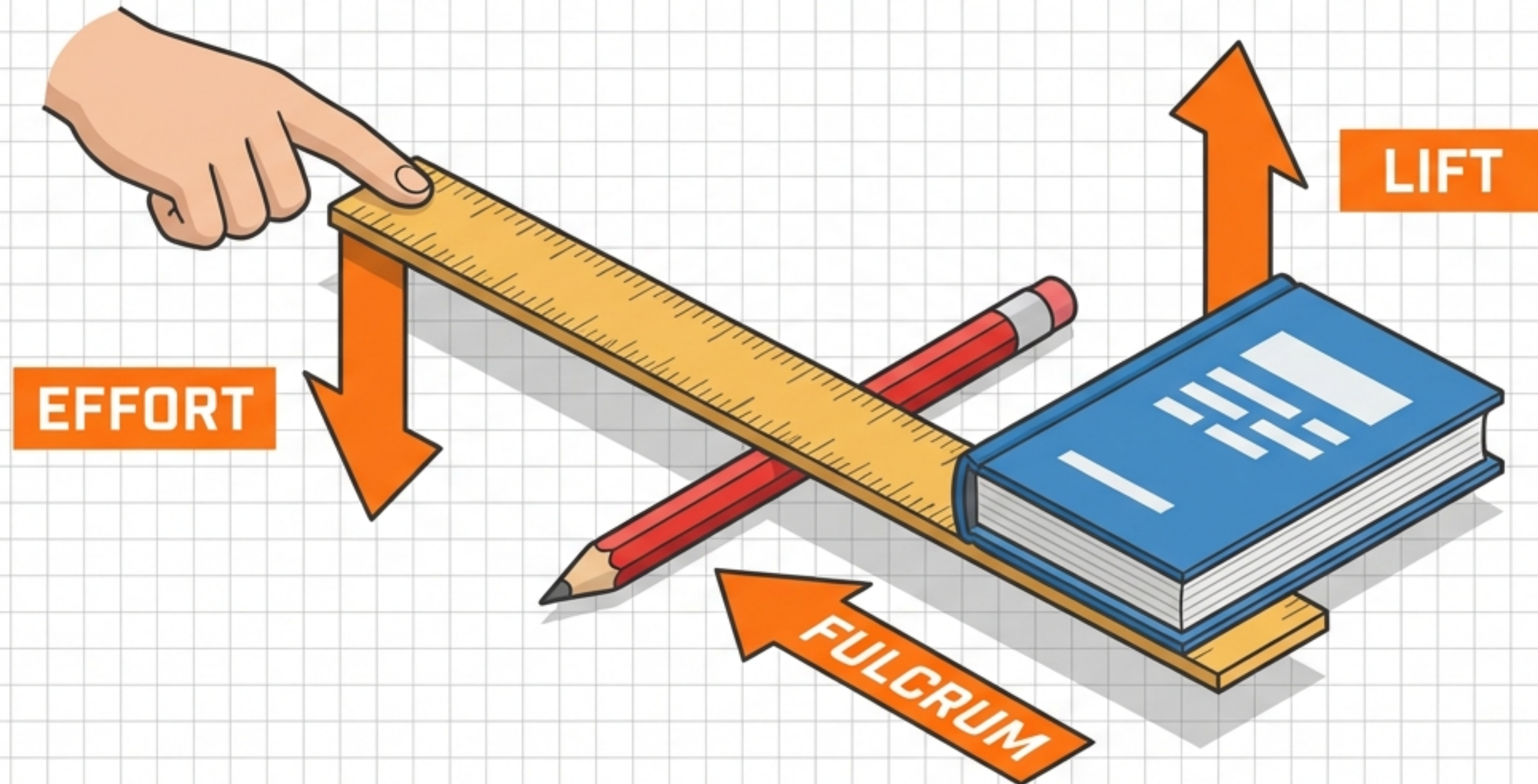
Class 3



Effort is between
Fulcrum and Load.

Activity: Build Your Own Lever

Can you make a lever using a ruler, a pencil, and a book?

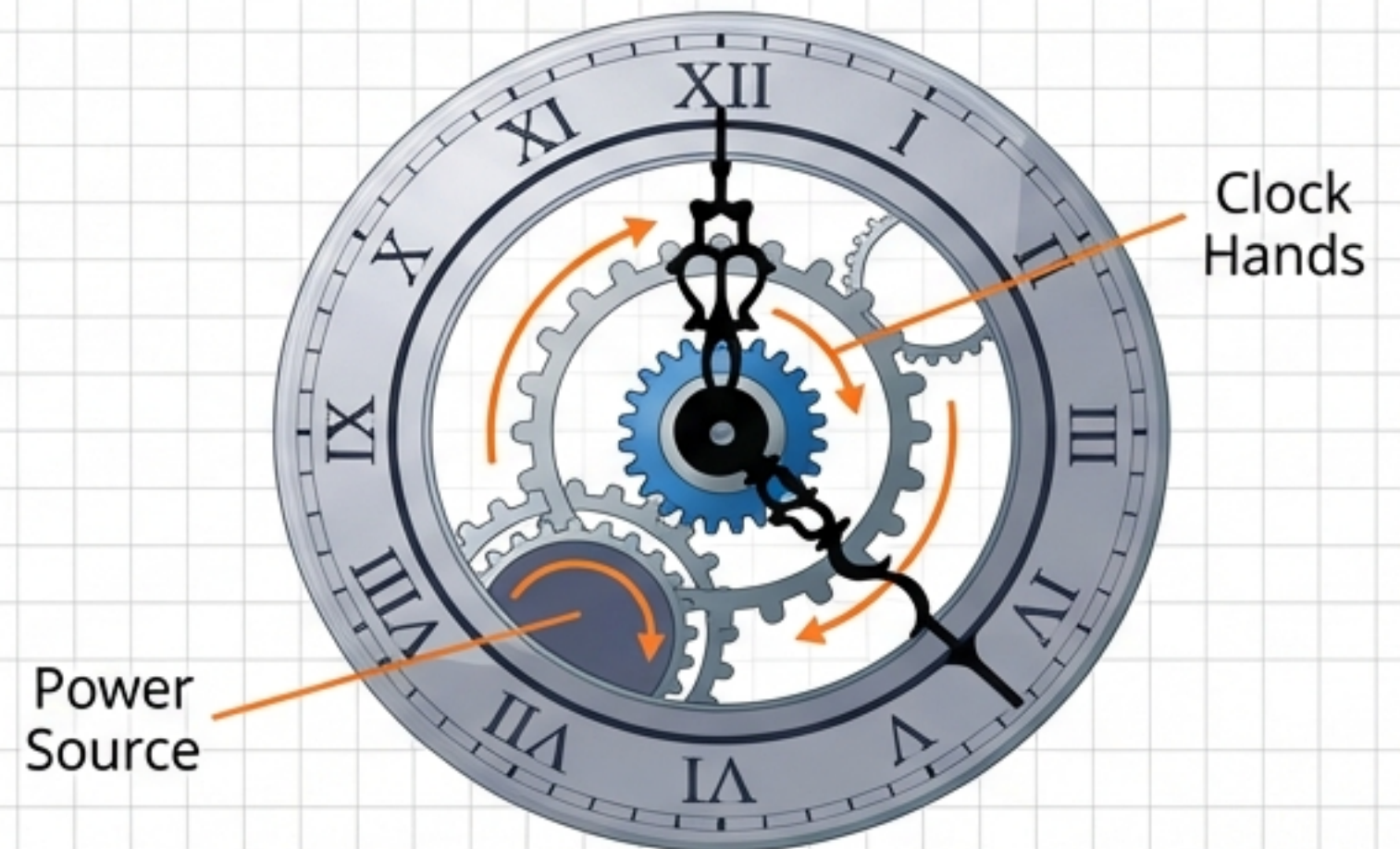
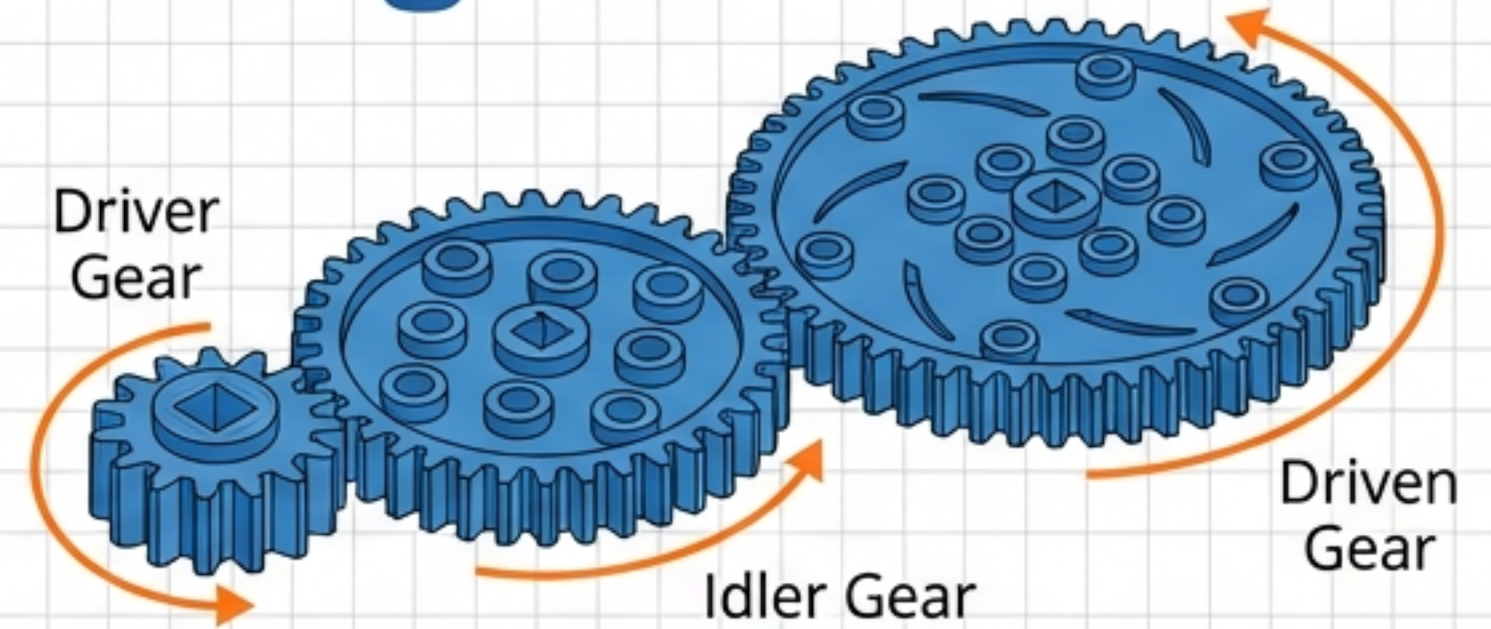


Gears: Keeping Things Moving

Mechanical parts with teeth that fit together to transmit force.

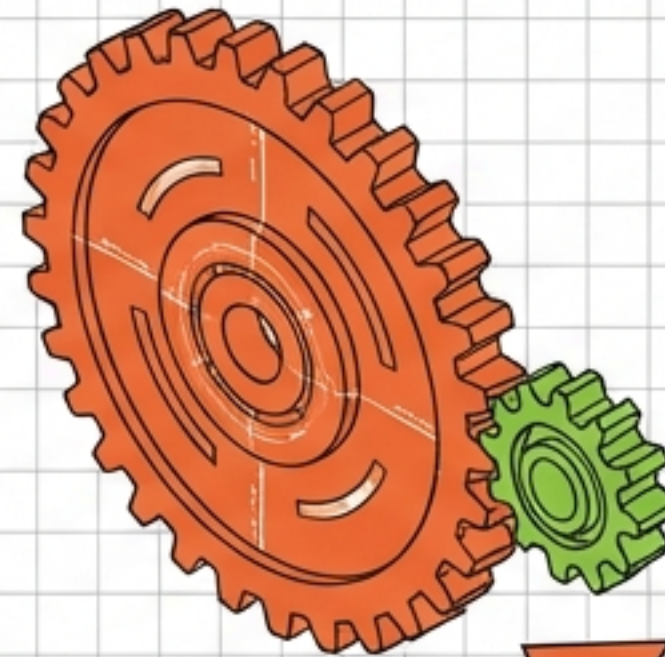
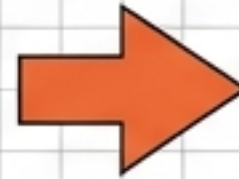
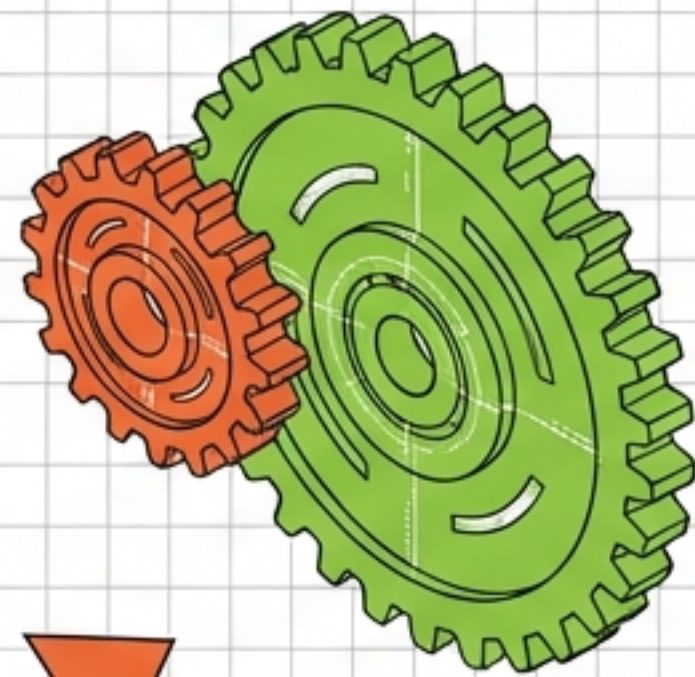
- **The 3 Jobs of a Gear**

- ➔ 1. Reverse direction of rotation.
- ⚙️ 2. Increase or decrease speed.
- ⚙️ 3. Transfer motion to a different axis.

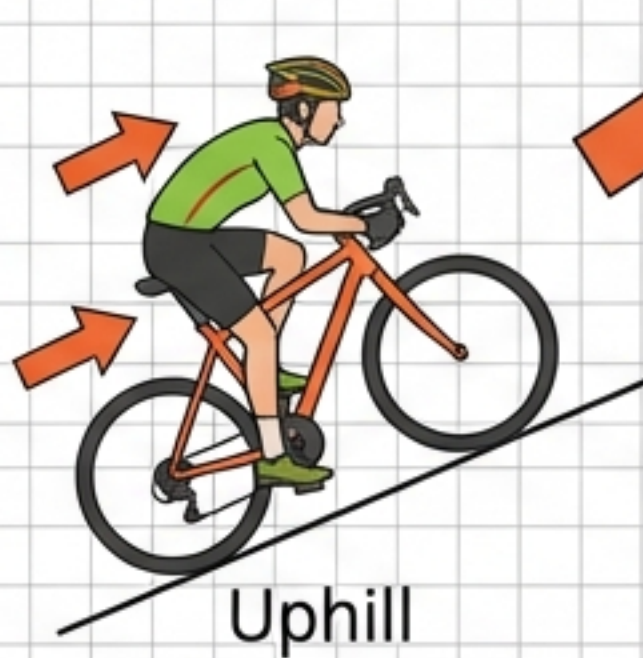


Torque vs. Speed

Small Gear (Driver)
+
Big Gear (Driven)
=
High Torque (Power)



Big Gear (Driver)
+
Small Gear (Driven)
=
High Speed



Uphill

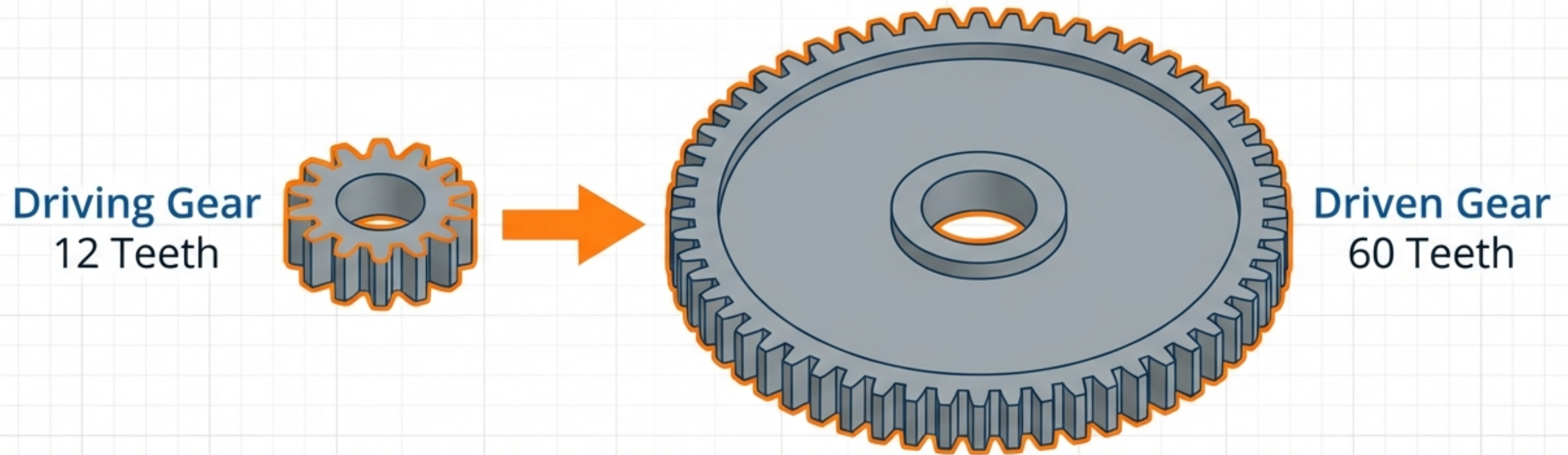


Downhill



Calculating Gear Ratio

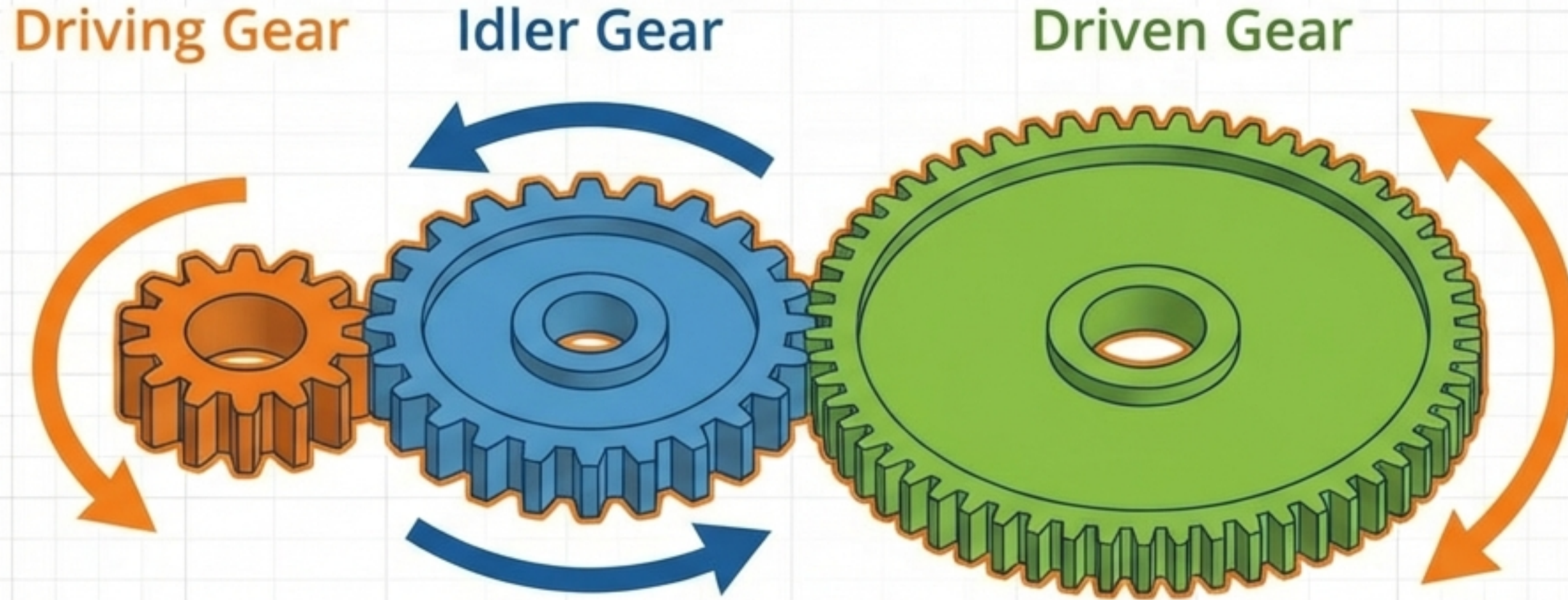
Gear Ratio = Driving Teeth : Driven Teeth



$$12 : 60 = 1 : 5 \text{ Ratio}$$

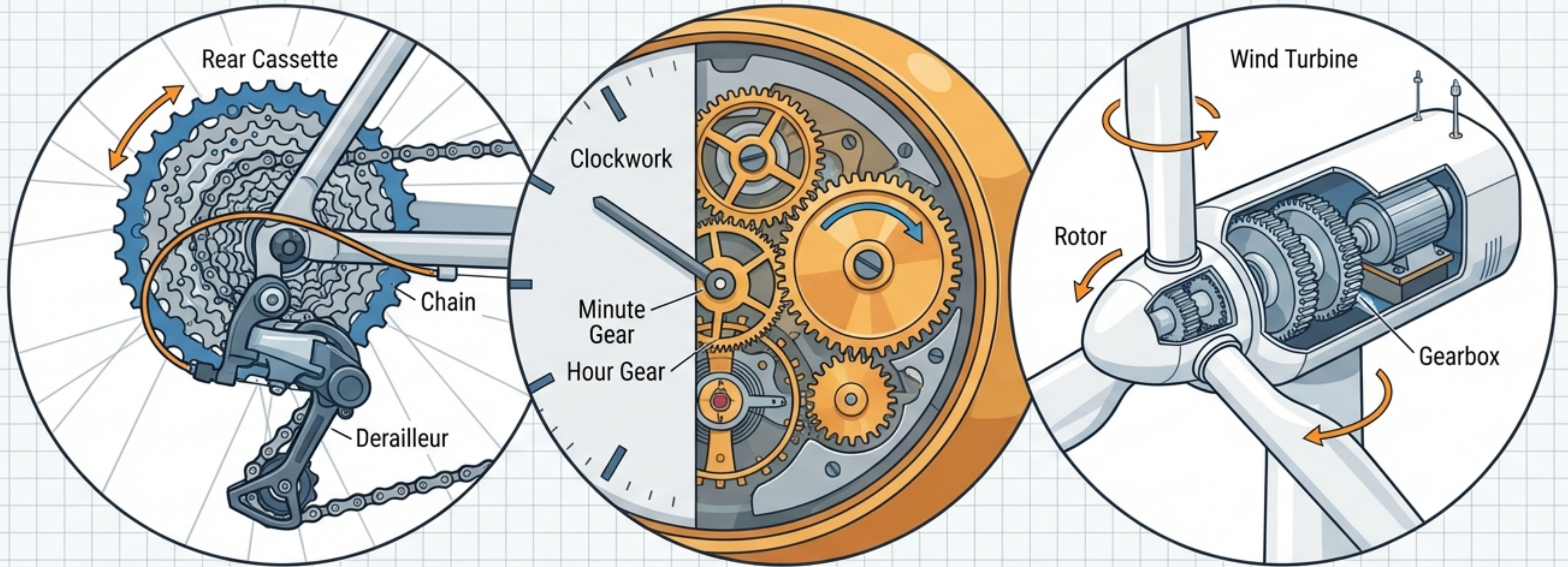
The small gear spins 5 times to turn the big gear once. This creates **POWER!**

The Idler Gear



- 1. Placed between driving and driven gears.
- 2. Changes direction of rotation.
- 3. NO effect on Speed or Torque.

Gears in the Real World



Can you name other machines that use gears? (Hint: Cars, Fans)

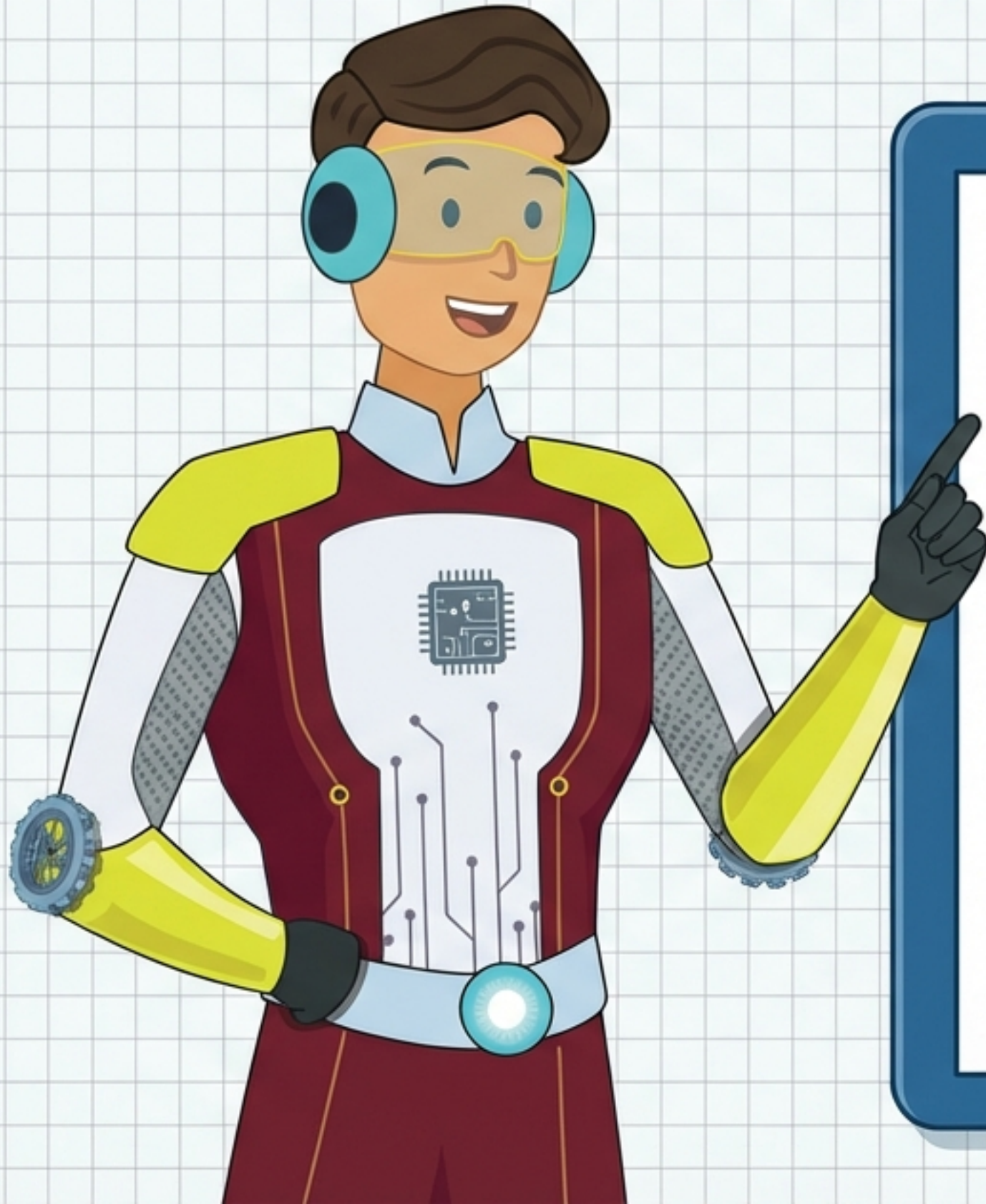
Tech Flash: Vocabulary

Tech Checklist

- ☒ **Structure:** Organization of elements.
- ☒ **Stability:** Ability to maintain form.
- ☒ **Balance:** Even distribution of weight.
- ☒ **Simple Machine:** Device to make work easier (Pulley, Lever, Wedge, Screw, Wheel & Axle, Inclined Plane).
- ☒ **Torque:** Force that causes rotation.



Engineering Challenge



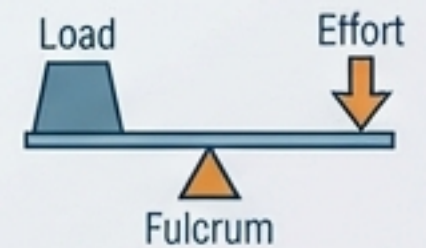
1. Why are bridges so strong?
(Hint: Structure & Stability)



2. Which bicycle needs more power:
Uphill or Downhill?



3. What do we call a lever where
the Load is in the middle?



You Are The Engineer

Engineers use knowledge to make lives better and use resources efficiently.

Keep exploring! Look for simple machines in your home today. What will you build next?

